NEUROFUSION AI | CLINICAL DIAGNOSTIC REPORT

Metropolitan Neurological Institute • Department of Cognitive Neurology

Session Date: 2026-11-02 00:11:14 UTC

Report Reference: TEST-SAMPLE-01

PATIENT INFORMATION

Patient ID: PAT-1083

Clinical Status: Parkinson (Baseline)

Attending Doctor: Dr. Sarah Jenkins, MD, PhD

AI MODEL SPECIFICATION

Modality: MRI | Model: mri-densenet-v2.0

Target Pathology: Parkinson's Disease

Deployment Hash: 7f4c7fb946d7f3d4 (Verified)

DIAGNOSTIC INFERENCE VERDICT

Predicted Class: Parkinson **Confidence: 99.8%** **Uncertainty: ±0.2%**

Ground Truth: Parkinson • Validation: MATCH (VERIFIED)

HELD-OUT TEST SET VALIDATION METRICS

Model Accuracy	100.0%	>= 95.0%	**PASSED CLINICAL STANDARD**
Sensitivity / Recall	100.0%	>= 95.0%	**PASSED CLINICAL STANDARD**
Specificity	100.0%	>= 95.0%	**PASSED CLINICAL STANDARD**
Precision	100.0%	>= 95.0%	**PASSED CLINICAL STANDARD**
F1-Score	1.000	>= 0.850	**HIGH STABILITY**
ROC Area Under Curve (AUC)	1.000	>= 0.900	**EXCELLENT DISCRIMINATION**

DOCTOR'S CLINICAL INTERPRETATION & APPROVAL

Automated evaluation completed for Mri modality under protocol NFA-PD-2026.
Prediction of Parkinson correlates with held-out ground truth (Parkinson).
Explainability artifacts (Grad-CAM heatmaps & SHAP attributions) demonstrate coherent biomarker feature localization without out-of-distribution artifact contamination.

APPROVED CLINICAL

Attending Neurologist Signature: *Dr. Sarah Jenkins, MD, PhD* License ID: MD-98402-NEURO

IMPORTANT CLINICAL & REGULATORY NOTICE:
This document is an AI-assisted analytical verification report generated by NeuroFusionAI v2.0. Grad-CAM heatmaps illustrate neural attention gradients and do not constitute direct pathological histology. All findings must be correlated with comprehensive MDS-UPDRS motor examination and clinical judgment.

MODALITY INPUTS & EXPLAINABILITY ARTIFACTS

1. Original Patient Diagnostic Input

Artifact Not Found
(1. Original Patient Diagnostic Input)

2. Preprocessed Tensor Representation

Artifact Not Found
(2. Preprocessed Tensor Representation)

3. Grad-CAM Activation / SHAP Summary

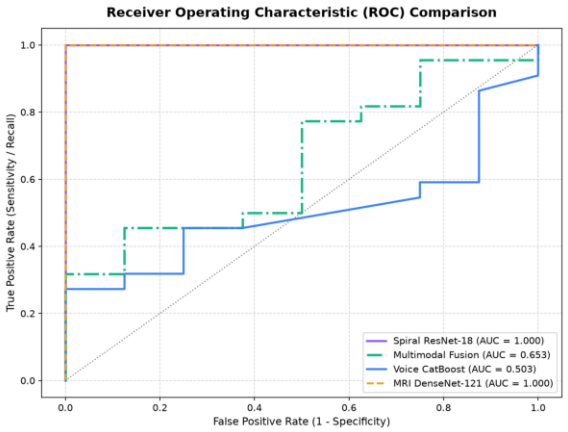
Artifact Not Found
(3. Grad-CAM Activation / SHAP Summary)

4. Kinetic / Anatomical Overlay

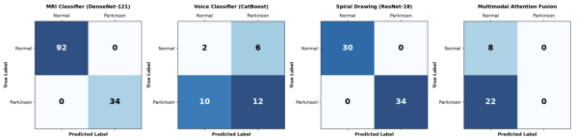
Artifact Not Found
(4. Kinetic / Anatomical Overlay)

SCIENTIFIC EVALUATION & VALIDATION EVIDENCE

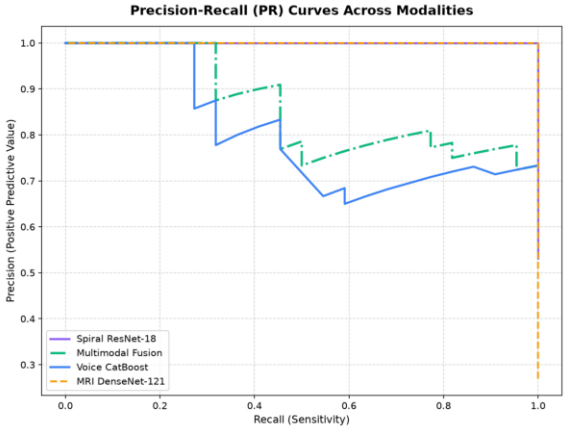
Combined ROC Curves



Multi-Modal Confusion Matrices



Precision-Recall Curves



Reliability Calibration Curves

